Cross-Site Transfer of Set-Based Deep Reinforcement Learning for EV Charging-Hub Control on Four Public Session Datasets

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***Abstract*—Deep reinforcement learning (DRL) controllers for electric-vehicle charging are usually trained and evaluated on data from a single site. This paper evaluates a proximal policy optimization (PPO) controller for a charging hub with a shared power limit at sites excluded from training, using 280 022 public AC sessions from four datasets (Palo Alto, Dundee, Perth and Kinross, Trondheim) and GB half-hourly prices in weekly episodes with chronological splits. The controller outputs a hub set-point that a safety layer allocates by least laxity; flat, DeepSets and self-attention state encoders were compared after tuning with Optuna. In leave-one-site-out tests with five seeds, the PPO agents had a mean regret with respect to the clairvoyant optimum of 7.1–7.5%, against 7.6% for least-laxity-first (LLF) and 3.6% for a receding-horizon linear program (MPC). With a 1 h error in the announced departures, the regret of MPC rose to 21.6%, that of PPO to 10.9–11.9% and that of LLF to 8.9%. The flat encoder had the lowest regret at the site with the shortest dwell times and the attention encoder at the residential site, and pooling three source sites did not reduce the regret at the held-out site with respect to the best single source. Data processing and experiment code are released.**

***Index Terms*—electric vehicles, smart charging, deep reinforcement learning, zero-shot generalization, set neural networks, smart grids**

I. Introduction

Uncoordinated charging of plug-in vehicles on a local scale can lead to grid problems, analysed in terms of power losses and voltage deviations, and coordinated charging was proposed to reduce them [1]. As the fleet grows, an intelligent charging management system is expected to be necessary to limit network reinforcement costs, voltage deviations and transformer saturation [2]. The time at which charging is shifted also matters for the equipment: simulations of a 25 kVA distribution transformer showed that temperature-based smart charging reduced the mean and variance of transformer aging, while delaying charging until after midnight could increase aging [3].

Optimization-based scheduling is already operated at scale: the Adaptive Charging Network uses an algorithm based on convex optimization and model predictive control and has been deployed at more than one hundred sites [4]. Such schedulers consume the departure time and energy announced by each driver [4], and the analysis of the public ACN-Data set found that these user inputs are unreliable and gave the worst prediction among the compared estimators [5].

Deep reinforcement learning (DRL) has been proposed as an alternative that formulates charging as a Markov decision process with unknown transition probabilities and learns a policy without a model of the system [6]. Continuous-action DRL has been applied to the charging of an individual vehicle under dynamic user behaviour and prices [7], and a review of DRL-based charging management lists the uncertainty of vehicle parameters, energy prices and loads as the main difficulty addressed by these methods [8]. Open simulators support this line of work: ACN-Sim replays ACN-Data sessions and exposes an OpenAI Gym interface [9], and the EV2Gym authors report that only a few smart charging simulators provide a Gym environment for RL [10].

Three limitations motivate this study. First, a DRL policy that is deployed at a new site faces a problem of zero-shot generalization, in which a policy must perform in situations not seen in training instead of overfitting to its training environment [11]; for charging, learning-based schedulers trained on pre-COVID data were reported to degrade after COVID [12]. Second, public session data come from different populations, such as a workplace campus [5] and residential apartment buildings [13], whose flexibility depends on parking and charger arrangements [14]. Third, DRL results vary with seeds and implementation details, so comparisons without repeated runs and significance tests are hard to interpret [15].

The objective of this paper is to measure, on four public session datasets from three countries, how a DRL controller for a charging hub with a shared power limit performs at a site that was excluded from training, and how this performance compares with heuristic, optimization-based and clairvoyant references when the announced departure times contain errors. The contributions are: (i) a harmonized benchmark of 280 022 public AC charging sessions with real half-hourly prices and temporally disjoint train, validation and test periods, replayed in a simulator checked against ACN-Sim [9]; (ii) a PPO con-

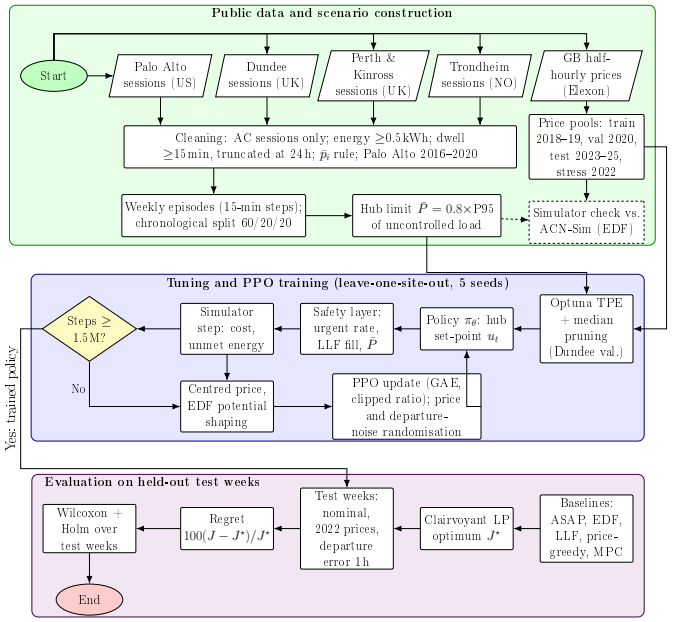


Fig. 1. Pipeline of the study: public data and scenario construction, hyperparameter tuning and PPO training with leave-one-site-out sources, and evaluation on held-out test weeks.

troller with a permutation-invariant encoder over the plugged-in vehicles and a one-dimensional set-point action that applies to hubs of any size; (iii) a leave-one-site-out evaluation with five seeds, paired statistical tests and ablations of the training design. The study falls within the ICIT topics of smart grids and of artificial intelligence for industrial informatics: charge point operators of public, workplace and fleet hubs face a contracted or transformer power limit and time-varying energy prices.

II. Data and Simulation Environment

Fig. 1 summarizes the pipeline; every table and figure is generated by the scripts cited.

A. Session datasets and cleaning

Four public datasets were used (Table I): the City of Palo Alto public charging network [16], the Dundee City Council public network [17], the Perth and Kinross Council network [18], and charging reports from Norwegian apartment buildings [13]. Each record was mapped to arrival time, departure (plug-out) time, delivered energy and charger identifier. Table I lists the cleaning rules and the sessions removed by each. Palo Alto sessions before 2016 were removed because the network was being built out (1 226 sessions in 2011 against 45 619 in 2019), the two Dundee 2024 files overlap, and DC rapid sessions leave no room for shifting. The maximum rate of session i is $\bar{p}_i = \min(\max(p^{\mathrm{nom}}, 1.1E_i/w_i), 22)$ kW, where w_i is the dwell time and p^{nom} is 6.6 kW for Palo Alto Level 2 (1.9 kW Level 1), 7.4 kW for Dundee and Perth and 3.7 kW for Trondheim; the observed average rate is a lower bound of the vehicle rate. The requested energy E_i is the delivered energy, capped at $0.98\,\bar{p}_i w_i$.

TABLE I
Sessions removed per cleaning rule, statistics after cleaning (idle share $1 - E_i/(\bar{p}_i w_i)$) and weekly episodes per split.

	Palo Alto (US, public)	Dundee (UK, public)	Perth (UK, public)	Trondheim (NO, resid.)
Raw records	259 415	501 450	66 664	6 878
Before 2016	−52 818	–	–	–
Duplicates	–	−110 957	−7	–
DC rapid / rapid models	–	−321 514	−51 435	–
Unparseable fields	0	−1 941	−58	−34
Dwell < 15 min	−8 068	−4 167	−1 652	−173
Energy < 0.5 kWh	−805	−511	−231	−14
Sessions kept	**197 724**	**62 360**	**13 281**	**6 657**
Dwell truncated at 24 h	22	2 457	459	456
Period	2016–20	2021–25	2016–19	2018–20
Chargers	80	67	40	110
Mean energy (kWh)	9.2	16.4	11.0	13.1
Mean dwell (h)	2.6	5.8	4.4	10.2
Mean idle share (%)	40	42	52	51
Weeks train/val./test	157/52/53	128/43/43	101/35/35	32/12/12
Median $\bar{P}$ train/test (kW)	80.2/40.3	56.7/53.1	18.0/30.9	13.5/39.9
Test weeks from	2019-12	2024-11	2019-02	2019-11

The sites differ in the behaviour that the transfer experiments rely on: the median arrival time is 17.1 h at the residential site (Trondheim) and 12.5–13.2 h at the public networks, and the median dwell is 10.3 h against 2.2–2.8 h.

B. Prices

Half-hourly GB market index prices (APX) for 2018–2025 were downloaded from the Elexon Insights API [19]; auctions with zero volume were treated as missing and interpolated, and the series was resampled to 15 min. The same price signal is used at every site so that differences between sites come from user behaviour and hub size. Price weeks are split into disjoint pools: training 2018–2019 (105 weeks), validation 2020 (52), test 2023–2025 (137) and a stress pool with the 2022 energy-crisis prices (52). Within an episode, prices are divided by their mean over the episode, and the controller sees the day-ahead prices of the next 24 h.

C. Episodes and hub limit

An episode is one calendar week of real arrivals (Monday 00:00 to Monday 00:00) followed by a 24 h tail, discretized in $\Delta = 15$ min steps (768 steps). Arrival steps are rounded up and departure steps down, with at least one step of dwell. Weeks with fewer than 10 sessions are discarded (3 weeks at Perth and Trondheim). The weeks of each site are split chronologically into 60% training, 20% validation and 20% test, so that test weeks are later than all training weeks (Table I). Week k of a split is paired with price week $k \bmod N$ of the corresponding pool; the pairing is fixed and identical for all methods. Each dataset is treated as one virtual hub: the sessions of all its chargers share one power limit, which represents an aggregator or a common feeder and not the physical layout of the network. The hub has $M = 48$ charger slots (the largest observed concurrency is 45; no session was lost to a full hub). Because the networks grew over the years

(Palo Alto multiplied its yearly sessions by 7 between 2012 and 2019), the hub limit $\bar{P}$ is set per week to 0.8 times the 95th percentile of the load that the week produces under uncontrolled charging, with a floor equal to the largest $\bar{p}_i$ of the week.

D. Objective and reference optimum

With rates $r_{i,t} \in [0, \bar{p}_i]$ while vehicle i is plugged in and $\sum_i r_{i,t} \le \bar{P}$, the weekly objective is

$$J = \sum_t p_t \, \Delta \sum_i r_{i,t} + \lambda \sum_i u_i, \qquad u_i = E_i - \Delta \sum_{t<d_i} r_{i,t}, \tag{1}$$

with normalized price p_t, undelivered energy $u_i \ge 0$ and $\lambda = 3$ (three times the mean price of the week). The clairvoyant optimum $J^\star$, which knows all arrivals, departures and energies of the week, is the linear program (1) solved with the HiGHS solver through SciPy. Results are reported as regret $100(J - J^\star)/J^\star$ (%), together with the fraction of requested energy delivered and the energy cost per delivered kWh.

E. Simulator and its check against ACN-Sim

The simulator is vectorized in NumPy over a batch of episodes. A new vehicle occupies the lowest free slot; at every step, rates are clipped to $\bar{p}_i$ and to the remaining energy and the hub limit is enforced (Section III-A). To check the implementation, five validation weeks per site were replayed with earliest-deadline-first (EDF) scheduling in this simulator and in ACN-Sim [9] (acnportal 0.3.3), using a 1000 V single-phase network (1 A = 1 kW), ideal batteries and a common maximum rate of 7 kW, since the EDF of ACN-Sim allocates up to the charger limit and not up to the vehicle limit. The difference in delivered energy, relative to demand, was between -0.26% and $+0.12\%$ over the 20 weeks (`scripts/validate_acnportal.py`).

III. Controller and Training

A. Action and safety layer

In a preliminary run on Dundee with one Beta-distributed action per slot (48 dimensions), the validation J per kWh reached 1.074 after 3×10^5 steps, against 1.081 for least-laxity-first and 0.923 for MPC with the safety layer; this variant is retained as an ablation. Since all vehicles pay the same price at a given step, the choice of *which* vehicle charges only affects the deadlines. The policy therefore outputs one hub set-point $u_t \in [0, 1]$, held for k simulator steps. A safety layer converts it into rates in two passes ordered by least laxity (laxity = announced time left minus time needed at $\bar{p}_i$): first, each vehicle receives the *urgent* rate $\underline{r}_{i,t} = [(e_{i,t} - \bar{p}_i(\hat{d}_i - t - 1)\Delta)/\Delta]^+$, the smallest rate that still allows completion at full power, up to $\bar{P}$; second, the remainder of $u_t\bar{P}$ is distributed, again by least laxity, up to each vehicle's feasible rate. The hub limit is never exceeded.

B. State and policy network

Each occupied slot is described by six features: presence, energy still needed expressed in days at $\bar{p}_i$, announced time left (days), laxity (days), $\bar{p}_i/22$ and remaining fraction of E_i. Global features (33 values) are sine and cosine of hour and weekday, occupancy, feasible power over $\bar{P}$, current price as z-score and rank within the next 24 h, the 24 hourly z-scores of the next day, elapsed fraction of the episode, urgent power over $\bar{P}$, the maximum deficit of the EDF demand condition (below) normalized by one day of $\bar{P}$, and pending energy normalized by one day of $\bar{P}$. Three encoders were compared: (a) a flat MLP over the 48 zero-padded slots and the global features; (b) DeepSets [20]: a shared MLP ϕ embeds each slot concatenated with the global features, followed by masked mean and max pooling; (c) the same encoder with one multi-head self-attention layer and layer normalization before pooling [21]. Actor and critic are one-hidden-layer heads on the pooled vector and the global features; the actor outputs the parameters $\alpha = 1 + \mathrm{softplus}(\cdot)$ and $\beta = 1 + \mathrm{softplus}(\cdot)$ of a Beta distribution for u_t, initialized with mean 0.6. Evaluation uses the mean of the Beta distribution. Encoders (b) and (c) are permutation invariant and do not depend on M.

C. Learning signal

Centred price. Since $\Delta \sum_{t,i} r_{i,t} = D - U$, with $D = \sum_i E_i$ exogenous and $U = \sum_i u_i$, (1) equals $\sum_t (p_t - 1)\Delta \sum_i r_{i,t} + (\lambda - 1)U + D$. Training on the first two terms changes the reward of each step but not the optimal policy. *Shaping.* The potential $\Phi(s) = -(\lambda - 1)\max\big(\sum_i [e_i - \bar{p}_i\tau_i\Delta]^+, \max_k [\sum_{\hat{d}_i \le \hat{d}_k} e_i - \bar{P}\tau_k\Delta]^+\big)$, with τ the announced steps left, is a lower bound on the energy of the plugged-in vehicles that can no longer be delivered (the second term is the EDF demand condition for the shared limit). The shaping term $\gamma\Phi(s') - \Phi(s)$ preserves the optimal policies [22]. *Normalization.* Global features are normalized with running mean and standard deviation (clipped to ± 10) and rewards are divided by the running standard deviation of the discounted return. *Augmentation.* In the final runs, each sampled training week is paired with a random week of the training price pool, and the announced departure of each vehicle is perturbed by $\mathcal{N}(0, \sigma^2)$, with $\sigma \sim U[0, 1.5]$ h drawn per batch of episodes (domain randomization [23]).

D. PPO, tuning and seeds

PPO [24] with generalized advantage estimation, Adam ($\epsilon = 10^{-5}$), gradient-norm clipping 0.5, value coefficient 0.5, 8 minibatches per epoch and linear decay of the learning rate was used. Hyperparameters were tuned with Optuna (TPE sampler, median pruner at half of the budget) [25] on the Dundee training weeks, scored on 20 Dundee validation weeks, with 10^6 simulator steps per trial and without augmentation (Table II). The DeepSets study ran 10 trials (7 completed, 3 pruned) and the MLP study 20 trials (13 completed, 7 pruned). DeepSets trials took between 14 and 76 min, whereas the first attention trial had not finished after 1.6 h; within the time available, the attention encoder was

TABLE II
SEARCH SPACE AND SELECTED HYPERPARAMETERS.

Parameter	Search space	DeepSets	MLP
Hidden width	{64, 128, 256}	64	128
Layers of ϕ / body	{1, 2, 3}	2	1
Activation	{tanh, ReLU}	tanh	tanh
Learning rate	$[10^{-4}, 10^{-3}]$ log	1.17×10^{-4}	1.43×10^{-4}
Entropy coef.	$[10^{-5}, 10^{-2}]$ log	9.7×10^{-3}	2.0×10^{-3}
γ (per decision)	{0.95, 0.97, 0.99}	0.97	0.95
GAE λ	{0.90, 0.95, 0.98}	0.90	0.98
Clip ratio	{0.1, 0.2, 0.3}	0.2	0.1
Epochs	{3, 5, 10}	10	10
Rollout length	{48, 96, 192}	192	48
Decision interval k	{2, 4} steps	2 (30 min)	2 (30 min)
Parallel weeks	{16, 32}	16	16
Validation J/kWh		0.9666	0.9697

therefore not tuned and reuses the DeepSets configuration with four heads. Final agents were trained for 1.5×10^6 simulator steps with 16 parallel weeks.

Seeds control the PyTorch initialization and sampling, NumPy, the sampling of training weeks, price weeks and departure noise (through a seeded generator), and the departure noise of the test conditions. Two runs with the same seed produced identical training rewards and test costs to the eighth decimal on the same machine. All runs used an Intel Core i7-1355U CPU (10 cores, 16 GB RAM, no GPU), Python 3.10, PyTorch 2.10, NumPy 1.26.4, SciPy 1.15.3, pandas 2.3.3 and Optuna 5.0.0.

IV. EXPERIMENTAL PROTOCOL

Baselines. ASAP charges every vehicle at its maximum rate and scales all rates proportionally when $\bar{P}$ is exceeded (uncontrolled charging). EDF and LLF fill $\bar{P}$ in order of earliest announced departure and of least laxity. Price-greedy books, at arrival, the cheapest known 15-min steps of each vehicle's window and charges at full power when the laxity reaches one step. MPC solves, at each arrival and at least every hour, the linear program (1) restricted to the vehicles plugged in, with day-ahead prices up to their announced departures, and applies the first step; it has no forecast of future arrivals. The +S variants apply the same safety layer as the PPO agent.

Experiments. E2 (leave-one-site-out): an agent is trained on the training weeks of three sites and tested on the test weeks of the fourth, for each encoder and five seeds (DeepSets and MLP) or three seeds (attention). E1 (single source): an agent trained on one site is tested on all four (two seeds, DeepSets and MLP). E3 (ablations, one seed, leave-one-site-out): no shaping, no centred price, per-slot actions, and no departure noise in training.

Test conditions. Nominal (test weeks with 2023–2025 prices), stress (same weeks with 2022 prices) and departure error (nominal, with the announced departure of every vehicle perturbed by $\mathcal{N}(0, 1\,\mathrm{h})$, seeded identically for all methods). The clairvoyant optimum uses the true departures.

Statistics. For each held-out site, regrets are averaged over seeds per test week and compared between methods with the two-sided Wilcoxon signed-rank test over weeks; p-values are adjusted with the Holm procedure over the four sites of each comparison.

TABLE III
REGRET (%) ON THE TEST WEEKS OF EACH HELD-OUT SITE (E2). PPO: MEAN ± STANDARD DEVIATION OVER SEEDS (5 SEEDS; ATTENTION 3 SEEDS). ASAP DOES NOT USE DEPARTURES AND IS UNCHANGED UNDER DEPARTURE ERROR.

	Method	Palo Alto	Dundee	Perth	Trondheim
Nominal	ASAP	5.9	18.5	11.6	20.5
	LLF	2.5	12.8	6.4	15.2
	Price-greedy+S	4.8	8.0	6.3	8.0
	MPC+S	4.2	3.8	3.4	1.1
	PPO-MLP	3.1±0.2	11.9±0.6	5.6±0.5	12.4±2.4
	PPO-DeepSets	4.3±0.3	11.7±0.7	5.9±0.1	11.3±1.7
	PPO-Attention	4.3±0.2	10.8±0.8	6.0±0.2	10.8±1.6
Dep. error	LLF	4.5	13.6	7.4	15.2
	Price-greedy+S	22.9	25.1	33.3	22.6
	MPC+S	21.5	19.0	28.3	11.2
	PPO-MLP	8.6±0.5	13.7±0.5	10.0±0.7	13.4±1.4
	PPO-DeepSets	11.6±0.6	13.7±0.4	9.6±0.4	12.2±1.3
	PPO-Attention	12.6±1.0	13.2±0.8	9.3±0.3	11.8±1.1

V. RESULTS

A. Held-out sites (E2)

Table III lists the regret on the test weeks of each held-out site and Fig. 2 shows its distribution over weeks. Under nominal conditions, MPC with the safety layer had the lowest mean regret at Dundee (3.8%), Perth (3.4%) and Trondheim (1.1%), and LLF the lowest at Palo Alto (2.5%). The three PPO encoders, trained without the held-out site, reached 7.1–7.5% averaged over the four sites (standard deviation over seeds 0.2–0.3 points), against 7.6% for LLF, 6.4% for price-greedy+S and 3.6% for MPC+S. Per site, PPO had a lower regret than LLF at Dundee, Perth and Trondheim (Holm-adjusted $p < 0.003$ for the three encoders) and a higher regret at Palo Alto, where PPO-MLP (3.1%) was also below MPC+S (4.2%, $p_{\mathrm{Holm}} < 0.001$). With the 2022 crisis prices, the regrets changed by at most 2.3 points and the method with the lowest regret at each site was the same as under nominal prices.

The components of the objective explain these positions (site averages). Under nominal conditions, MPC+S paid 0.966 times the mean price per delivered kWh and delivered 97.2% of the requested energy, LLF paid 1.034 and delivered 98.9%, and ASAP paid 1.034 and delivered 96.6%. The PPO agents delivered 98.2–98.5% at 1.018–1.023: a delivered fraction near that of LLF and a smaller part of the price reduction obtained by MPC.

B. Errors in the announced departure

When the announced departures were perturbed by $\mathcal{N}(0, 1\,\mathrm{h})$, the mean regret of MPC+S rose from 3.6% to 21.6% and that of price-greedy+S from 6.4% to 26.1%, while LLF rose from 7.6% to 8.9% and the PPO agents from 7.1–7.5% to 10.9–11.9% (Table III). The delivered fraction fell to 88.1% for MPC+S and 86.9% for price-greedy+S,

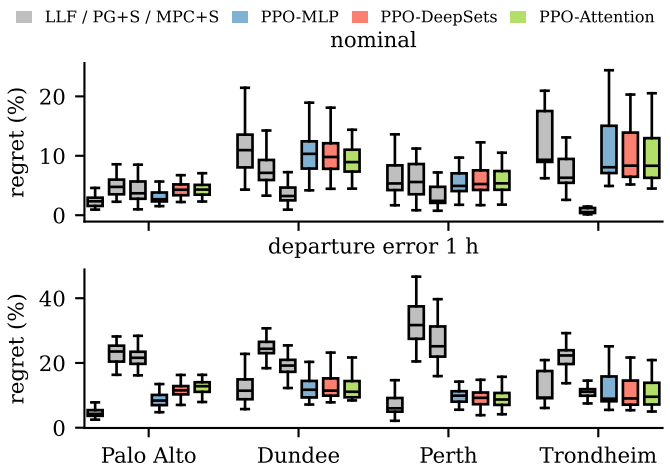


Fig. 2. Distribution of the weekly regret on held-out sites (E2). Grey boxes, from left to right: LLF, price-greedy+S and MPC+S. PPO boxes use the seed-averaged regret of each week.

against 95.6–96.5% for PPO and 98.2% for LLF: a plan that finishes each vehicle at its announced departure leaves energy undelivered whenever the vehicle leaves earlier. All three PPO encoders had a lower regret than MPC+S at Palo Alto, Dundee and Perth ($p_{\text{Holm}} < 10^{-4}$), with no difference detected at Trondheim ($p_{\text{Holm}} \geq 0.52$). Compared with LLF, PPO had a higher regret at Palo Alto and Perth ($p_{\text{Holm}} < 10^{-7}$), no difference detected at Dundee ($p_{\text{Holm}} \geq 0.30$) and a lower regret at Trondheim ($p_{\text{Holm}} \leq 0.01$). The size of this effect is consistent with the unreliability of user-provided departure times reported for ACN-Data [5], and with the use of departure-noise randomization during training [23]; the contribution of that randomization is isolated in the ablation below.

C. Encoders and transfer gap

Differences between encoders depended on the site. At Palo Alto, the site with the shortest dwell times (Table I), the flat MLP had a lower regret than both set encoders (median difference 1.0–1.1 points, $p_{\text{Holm}} < 10^{-8}$); at Trondheim, the residential site, the attention encoder had a lower regret than the MLP (10.8% vs 12.4%, $p_{\text{Holm}} = 0.012$). Within the E2 runs, each site is a training source in three of the four configurations, which gives a measure of the transfer gap: the regret on a site when it was held out minus the regret on the same site when it was a source averaged 0.5 (DeepSets), 0.6 (attention) and 0.8 (MLP) points, with the largest gap at Dundee (1.5–2.4 points).

Fig. 3 reports the single-source experiment (E1, two seeds). For DeepSets, the mean regret on the diagonal (trained and tested on the same site) was 6.1% and off the diagonal 7.5%; for the MLP, 5.5% and 5.8%. Training on Palo Alto, the site with the shortest dwell times, gave the highest regrets on Dundee and Trondheim for both encoders (DeepSets 13.1% and 14.6%). For Trondheim, the best single source (MLP trained on Perth, 7.8%; DeepSets trained on Dundee, 9.0%)

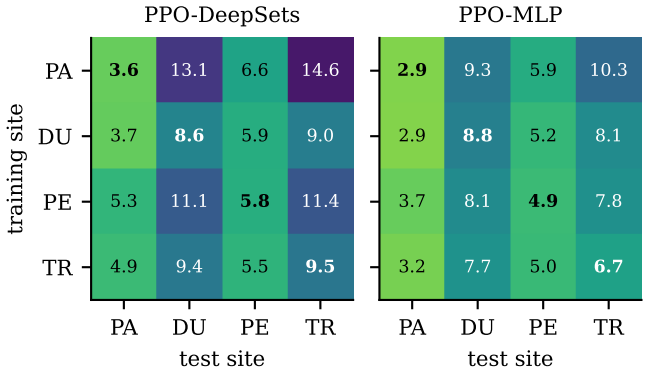


Fig. 3. Single-source transfer (E1): mean regret (%) on the test weeks of each site (columns) for agents trained on one site (rows), two seeds; diagonal in bold.

TABLE IV
ABLATIONS (E3): DEEPSETS, LEAVE-ONE-SITE-OUT, SEED 0; MEAN OVER THE FOUR HELD-OUT SITES. DELIVERED ENERGY AND COST ARE UNDER NOMINAL CONDITIONS.

Variant	Regret (%) Nominal	Regret (%) Dep. error	Delivered (%)	Cost
Full configuration	7.2	11.4	98.2	1.019
No centred price	10.0	19.7	96.9	1.021
No shaping	7.6	12.2	98.2	1.022
One action per slot	7.8	17.2	98.1	1.022
No departure noise in training	6.3	17.0	97.7	1.001

had a lower regret than the corresponding E2 agents trained on three sources (12.4% and 11.3%), so pooling sources did not by itself reduce the regret at the held-out site. With three sources the 16 parallel training weeks are shared among sites; this factor was not isolated, and E1 used two seeds.

D. Ablations (E3)

Table IV compares the DeepSets agent (seed 0) with four variants, each removing one element of the training design, over the 143 test weeks of the held-out sites (paired Wilcoxon tests against the full configuration). The nominal regret rose without the centred price ($p = 3 \times 10^{-10}$; at Trondheim from 10.3% to 23.5%), without shaping ($p = 6 \times 10^{-8}$) and with one action per slot ($p = 9 \times 10^{-6}$). Training without departure noise lowered the nominal regret (median difference -0.34 points, $p = 0.014$) and raised the regret under departure error ($p = 9 \times 10^{-17}$), the trade-off that domain randomization targets [23].

E. Example week and computation time

Fig. 4 shows the first four days of a Dundee test week (Dundee held out). MPC+S concentrated charging in the low-price hours, alternating between zero and $\bar{P}$; LLF charged at $\bar{P}$ whenever vehicles were present; PPO-DeepSets lowered the set-point in high-price hours while the urgent rates kept charging active. Over this week, the energy cost per delivered kWh was 1.038 (LLF), 0.955 (MPC+S) and 1.018 (PPO), and the delivered fraction 99.8%, 98.6% and 99.7%. On one CPU thread (`scripts/timing.py`), a PPO decision including

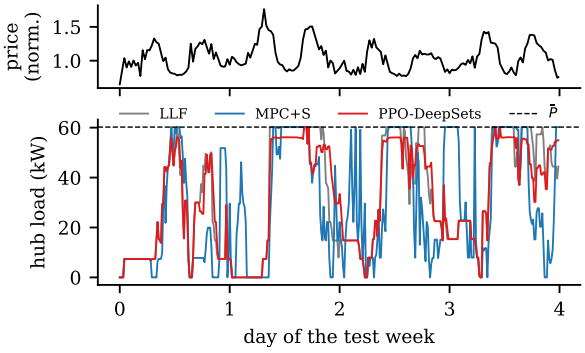


Fig. 4. Normalized price and hub load during four days of a Dundee test week with Dundee held out of training (PPO-DeepSets, seed 0).

the safety layer took 0.6–0.9 ms (MLP), 1.1–1.7 ms (DeepSets) and 1.4–1.8 ms (attention), an MPC+S step 0.9–2.6 ms on average (maxima 13–31 ms) and an LLF step 0.03 ms, all far below the 15-min control interval.

For a charge point operator with a contracted power limit, two cases follow from these results. Where drivers announce departures through an app without incentives to be accurate [5], MPC+S left 11.9% of the energy undelivered in the departure-error test, the PPO agents 3.5–4.4% and LLF 1.8%. Where departures are known, as in a fleet depot with fixed shifts, MPC+S had the lowest regret at three of the four sites.

VI. Conclusion

Trained on three sites and tested on the fourth, the PPO controller had a mean regret 0.1–0.5 points below LLF and 3.5–3.9 points above MPC under nominal conditions, and 2.0–3.1 points above LLF and 9.6–10.7 points below MPC with a 1 h departure error. The encoder with the lowest regret depended on the site, and the transfer gap averaged 0.5–0.8 points. The choice between optimization and a learned policy therefore depends on how reliable the announced departures are [5].

The study has limitations. One GB price series was used for all sites; each dataset was treated as one virtual hub with a single limit, without phase constraints; the delivered energy was taken as the requested energy and the departure error was synthetic; the hub limit and λ were fixed; the attention encoder was not tuned; E1 and E3 used two and one seeds; training was limited to 1.5×10^6 steps on a laptop CPU; and no field test was performed. Future work includes site-specific prices, per-location hubs with three-phase limits as in ACN-Sim [9], MPC with arrival forecasts, policies that combine a learned component with the optimization plan, vehicle-to-grid operation [10], and a field evaluation. The datasets are public [13], [16]–[19]; the code that produces every table and figure of this paper is available at [repository URL].

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